

Answer Key: Remote Sensing C Test

1 - C	26 - D
2 - B	27 - B
3 - B	28 - B
4 - C	29 - C
5 - B	30 - B
6 - B	31 - C
7 - C	32 - B
8 - B	33 - C
9 - C	34 - C
10 - C	35 - B
11 - B	36 - B
12 - C	37 - C
13 - C	38 - C
14 - C	39 - B
15 - C	40 - A
16 - B	41 - B
17 - B	42 - B
18 - C	43 - A
19 - C	44 - B
20 - C	45 - C
21 - B	46 - B
22 - C	47 - B
23 - B	48 - B
24 - C	49 - B
25 - B	50 - C

Section 6:

1. 2 pts define, 1 pt positive
 2. 2 pts contrast, 2 pts example
 3. 2 pts presence, 1 pt albedo
 4. 2 pts each part
 5. 2 pts explain, 2 pts ratio
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1. RF is the change in energy flux in the atmosphere (W/m^2). Positive RF = Warming.
 2. Rayleigh ($x \ll 1$). Particles like N_2 or O_2 molecules. Mie ($x \approx 1$) Larger particles like aerosols, dust, or water droplets.
 3. They increase the stratospheric albedo (direct effect) by scattering incoming solar radiation back to space, causing a net cooling effect.
 4. $E = \sigma T^4 = 459 \text{ W/m}^2$. Actual flux = $\epsilon \sigma T^4 = 413 \text{ W/m}^2$
 5. Methane has a much higher GWP over 20 years ($\sim 80\times$) than 100 years ($\sim 28\times$). This is because Methane has a much shorter atmospheric lifetime (~ 12 years) compared to CO_2 ; its impact is front-loaded.

Section 7:

1. 3pts all, 1 pt if only only cloud discussed
 2. 2 pts each
 3. 4 pts all
 4. 2 pts each part
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1. (3 pts) Cirrus: High/thin, net warming (high greenhouse effect, low albedo). Stratus: Low/thick, net cooling (high albedo, low greenhouse effect).
 2. (4 pts) Higher dBZ indicates larger particles (like rain drops or ice crystals) or higher concentrations. It helps distinguish between light fog and heavy precipitation. Y-axis is height (km)
 3. (4 pts) The two frequencies (Ka and Ku bands) have different attenuation rates when passing through liquid vs. ice. By comparing the signal decay, the DPR can identify the "bright band" (melting layer).
 4. (4 pts) AOD is a measure of how much light is blocked by aerosols. High AOD decreases surface solar radiation (cooling the surface) but can warm the atmosphere if the aerosols are absorbing (like soot).

Section 8:

1. 4 pts all
2. 3 pts all
3. 3 pts all
4. 1 pt each

1. $T = 2\pi \sqrt{r^3/GM}$. $R = 6371 + 700 = 7071$ km. $T = 2\pi \sqrt{(7171000^3/3.986 \cdot 10^{14})} = 5925s = 98.7$ min
2. It ensures the satellite passes over a spot at the same local solar time, keeping illumination (sun angle/shadows) constant, which is vital for comparing images over time.
3. Decreasing altitude increases the revisit period (the satellite moves faster and the swath width usually narrows, requiring more orbits to cover the same area).
4. Spatial: Size of the smallest detectable object (pixel size). Temporal: How often the sensor records data for the same location.

Section 9:

1. 2 pts each
2. 3pts all
3. 2 pts each
4. 4 pts all
5. 2 pts each

1. Phytoplankton fix CO_2 into organic matter via photosynthesis. When they die, they sink to the deep ocean, effectively "pumping" carbon out of the atmosphere for long-term storage.
2. It is the large-scale circulation that moves air from the tropical troposphere up into the stratosphere and then poleward/downward. It transports O_3 and N_2O .
3. Higher CO_2 lowers ocean pH, reducing the availability of carbonate ions needed for shells $CaCO_3$. Remote sensing can monitor this via changes in particulate inorganic carbon (PIC) or sea surface salinity/temp proxies.
4. $\Delta F = 5.35 \ln(2) = 3.71$ W/m². $\Delta T = \lambda * \Delta F = .8 * 3.71 = 2.97K$ (or °C)
5. Rice cultivation, livestock (enteric fermentation), landfills, or natural gas leaks.

Section 10:

1. 6 pts all

1. $T_e = [((1 - \alpha) * S_0) / 4\sigma]^{1/4}$. $T_e = [((1 - .3) * 1361) / 4 * (5.67 * 10^{-8})]^{1/4} = 255K$ or $-18^\circ C$